

HKUST MATH 2431 - problem set 5

1.

1. (*) σ -algebras and random variables

[4 Points]

In a random experiment, a green and a blue die are tossed at the same time. We choose $\Omega = \{1, \dots, 6\}^2$ (with (ω_1, ω_2) denoting the numbers shown by the (blue, green) die) and consider the following two σ -algebras:

$$\mathcal{F} = \mathcal{P}(\Omega),$$

$$\mathcal{F}_{\text{sym}} = \{A \subseteq \Omega : \text{for all } (\omega_1, \omega_2) \in \Omega, (\omega_1, \omega_2) \in A \Leftrightarrow (\omega_2, \omega_1) \in A\}.$$

- (a) Note that $\mathcal{F}_{\text{sym}}$ contains only sets that are symmetric with respect to the exchange of the order of ω_1 and ω_2 . In which situations does it make sense to use $\mathcal{F}_{\text{sym}}$ to model a probability space? Give a concrete example.

1.a. Need a situation where the two dice are considered as indistinguishable. For example, the number of 1s/2s/3s/4s/5s/6s rolled, in which the identities of the dice do not matter.

- (b) Show that $\mathcal{F}_{\text{sym}}$ is indeed a σ -algebra.

1.b. - $\Omega \in \mathcal{F}_{\text{sym}}$, as it contains all possible pairs (w_1, w_2) and thus the swapped pairs as well.

- For any $A \in \mathcal{F}_{\text{sym}}$ $A^c \in \mathcal{F}_{\text{sym}}$

$$(w_1, w_2) \in A \Leftrightarrow (w_2, w_1) \in A$$

$$\Rightarrow (w_1, w_2) \notin A \Leftrightarrow (w_2, w_1) \notin A$$

$$\Rightarrow (w_1, w_2) \in A^c \Leftrightarrow (w_2, w_1) \in A^c$$

$$\Rightarrow A^c \in \mathcal{F}_{\text{sym}}$$

- For any $(A_i \in \mathcal{F}_{\text{sym}})_{i \in \mathbb{N}}$, $\bigcup_{i \in \mathbb{N}} A_i \in \mathcal{F}_{\text{sym}}$

$$(w_1, w_2) \in \bigcup_{i \in \mathbb{N}} A_i$$

$$\Rightarrow (w_1, w_2) \in A_k \text{ for some } k \in \mathbb{N}$$

$$\Rightarrow (w_2, w_1) \in A_k$$

$$\Rightarrow (w_2, w_1) \in \bigcup_{i \in \mathbb{N}} A_i$$

Similarly for the reverse direction.

(c) Consider the two subsets of Ω :

$A =$ "one die shows the number 3",

$B =$ "the blue die shows the number 3".

Show that $A \in \mathcal{F}_{\text{sym}}$ but $B \notin \mathcal{F}_{\text{sym}}$.

1.c. $A = \{(\omega_1, \omega_2) \in \Omega : (\omega_1 = 3 \text{ and } \omega_2 \neq 3) \text{ or } (\omega_1 \neq 3 \text{ and } \omega_2 = 3)\}$

Consider the substitution $\omega_1 \mapsto \omega_2$ and $\omega_2 \mapsto \omega_1$ in the predicate.

$$A' = \{(\omega_1, \omega_2) \in \Omega : (\omega_2 = 3 \text{ and } \omega_1 \neq 3) \text{ or } (\omega_2 \neq 3 \text{ and } \omega_1 = 3)\}$$

The predicate in A' is equivalent to that in A

that is $A' = A$.

So $(\omega_1, \omega_2) \in A \Leftrightarrow (\omega_2, \omega_1) \in A$, thus $A \in \mathcal{F}_{\text{sym}}$

$$B = \{(\omega_1, \omega_2) \in \Omega : \omega_1 = 3\}$$

Counterexample: $(3, 1) \in B$ but $(1, 3) \notin B$

thus $B \notin \mathcal{F}_{\text{sym}}$

(d) Consider the two maps

$$X : \begin{cases} \Omega \rightarrow \mathbb{R} \\ (\omega_1, \omega_2) \mapsto \omega_1 \end{cases}$$

“number shown by the blue die”

$$S : \begin{cases} \Omega \rightarrow \mathbb{R}, \\ (\omega_1, \omega_2) \mapsto \omega_1 + \omega_2 \end{cases}$$

“sum of numbers shown by both dice”

Which of these is/are a random variable/random variables on $(\Omega, \mathcal{F}_{\text{sym}})$?

1. d. X is not a RV on $(\Omega, \mathcal{F}_{\text{sym}})$

Consider event $\{X=6\}$ ($\{6\}$ is Borel)

$$X^{-1}(\{6\}) = \{(6,1), (6,2), (6,3), \\ (6,4), (6,5), (6,6)\}$$

Note $(6,1) \in X^{-1}(\{6\})$ but $(1,6) \notin X^{-1}(\{6\})$

So $\{6\}$ is measurable but $X^{-1}(\{6\})$ is not,

so X is not a measurable func, so not a RV.
with resp. to $(\Omega, \mathcal{F}_{\text{sym}})$

S is a RV on $(\Omega, \mathcal{F}_{\text{sym}})$

As Ω is countable, image of S is countable.

As S is discrete, it suffices to consider the singleton sets on $\mathbb{R}$.

$$S(\Omega) = \{2, \dots, 12\}$$

Consider the singleton set $\{n\}$

$$S^{-1}(\{n\}) = \{(w_1, w_2) \in \Omega : w_1 + w_2 = n\}$$

Under the substitution $w_1 \mapsto w_2, w_2 \mapsto w_1$ in the predicate

$$S^{-1}(\{n\}) = \{(w_1, w_2) \in \Omega : w_2 + w_1 = n\}$$

The predicates are equivalent as $+$ is commutative.

$$\text{So } S^{-1}(\{n\}) = S^{-1}(\{n\})$$

$$\text{So } (w_1, w_2) \in S^{-1}(\{n\}) \Leftrightarrow (w_2, w_1) \in S^{-1}(\{n\})$$

So S is measurable with resp. to $(\Omega, \mathcal{F}_{\text{sym}})$

(as we have checked the singleton images)

4.

4. (*) Medians of distribution functions

[4 Points]

A real number m is called a *median* of the distribution function F whenever

$$(*) \quad \lim_{y \uparrow m} F(y) \leq \frac{1}{2} \leq F(m).$$

- (a) Show that (*) is equivalent to the requirement $\mathbf{P}([m, \infty)) \geq \frac{1}{2}$ and $\mathbf{P}((-\infty, m]) \geq \frac{1}{2}$ for the probability measure $\mathbf{P}$ associated with F .

4.a. $\lim_{y \uparrow m} F(y) \leq \frac{1}{2}$ and $\frac{1}{2} \leq F(m)$

$\Rightarrow \lim_{y \uparrow m} \mathbf{P}((-\infty, y]) \leq \frac{1}{2}$ $\Rightarrow \frac{1}{2} \leq \mathbf{P}((-\infty, m])$

$\Rightarrow \lim_{y \uparrow m} (1 - \mathbf{P}((-\infty, y])) \geq \frac{1}{2}$ $\Rightarrow \mathbf{P}([-\infty, m]) \geq \frac{1}{2}$

$\Rightarrow \lim_{y \uparrow m} \mathbf{P}((y, \infty)) \leq \frac{1}{2}$

$\Rightarrow \mathbf{P}([m, \infty)) \geq \frac{1}{2}$

Take a sequence $(y_i)_{i \in \mathbb{N}}$ that is increasing with $y_i < m$ and $y_i \uparrow m$ as $i \rightarrow \infty$, then (y_i, ∞) is decreasing and converges to $\cap (y_i, \infty) = [m, \infty)$

- (b) Show that every distribution function F has at least one median.

Hint: Consider $\alpha = \sup\{x \in \mathbb{R} : F(x) < \frac{1}{2}\}$, show that α satisfies (*).

4b. $\alpha := \sup \{x \in \mathbb{R} : F(x) < \frac{1}{2}\}$

α exists, as $F(\infty^-) = 1$ and we can make $\varepsilon > 0$ small enough such that $F(x) > 1 - \varepsilon \geq \frac{1}{2}$ for large enough x which by

monotonicity of F means $x \geq y$ for all $F(y) < \frac{1}{2} < F(x)$,
 Now we show $F(\alpha) \geq \frac{1}{2}$ and by LUB prop of $\mathbb{R}$,
 If $F(\alpha) < \frac{1}{2}$, then $\alpha \in \{x \in \mathbb{R} : F(x) < \frac{1}{2}\}$ a least upper bound exist.

and $\alpha \geq y$ for all $y \in \{x \in \mathbb{R} : F(x) < \frac{1}{2}\}$ ①

Now, as F is right continuous, we can

find $F(\alpha) \leq F(\alpha + \delta) < \frac{1}{2}$ for small enough $\delta > 0$

which contradicts with ① as $\alpha + \delta \in \{x \in \mathbb{R} : F(x) < \frac{1}{2}\}$

So $F(\alpha) \geq \frac{1}{2}$.

Now we show $\lim_{y \uparrow \alpha} F(y) \leq \frac{1}{2}$. By 4.9, the limit exists.

for any sequence $(y_i)_{i \in \mathbb{N}}$ where $y_i < \alpha$

and $y_i \uparrow \alpha$ as $i \rightarrow \infty$

as $y_i < \alpha$, $y_i \in \{x \in \mathbb{R} : F(x) < \frac{1}{2}\}$

So $F(y_i) < \frac{1}{2}$ ②

Assume $\lim_{y \uparrow \alpha} F(y) > \frac{1}{2}$

We can

make $\varepsilon > 0$

small enough such that for large enough i ,

$$F(y_i) \geq \lim_{y \uparrow \alpha} F(y) - \varepsilon > \frac{1}{2},$$

which contradicts with ②.

$$\text{So } \lim_{y \uparrow \alpha} F(y) \leq \frac{1}{2}.$$

So α is a median.

(c) Show that the set of medians of F is a closed interval of $\mathbb{R}$.

Hint: Consider $\beta = \sup\{x \in \mathbb{R} : F(x) \leq \frac{1}{2}\}$, show that the median fulfills $m \in [\alpha, \beta]$.

4.c. $\beta = \sup \{x \in \mathbb{R} : F(x) \leq \frac{1}{2}\}.$

For similar reason to α , β exists.

We see $\{x \in \mathbb{R} : F(x) < \frac{1}{2}\} \subseteq \{x \in \mathbb{R} : F(x) \leq \frac{1}{2}\}$

So any upper bound of the latter is also an upper bound of the former, so $\beta \geq \alpha$, so $[\alpha, \beta]$ is nonempty.

Assume $m < \alpha$. As α is the least upper bound of $\{x \in \mathbb{R} : F(x) < \frac{1}{2}\}$, m is in the set, so $F(m) < \frac{1}{2}$, which is not a median.

We show $m \in [\alpha, \beta]$ is a median and $m > \beta$ is not a median. α is already known to be a median.

For any $m \geq \alpha$, F is monotone, so $F(m) \geq F(\alpha) \geq \frac{1}{2}$.

Now we show for any $m \leq \beta$, β is a median by additionally showing $\lim_{y \uparrow m} F(y) \leq \frac{1}{2}$.

Turns out the same proof in 3.b. works, just with $F(x) > \frac{1}{2}$ replaced by $F(x) \geq \frac{1}{2}$.

So $m \in [\alpha, \beta]$ is a median.

Finally for any $x > \beta$, $F(x) \geq F(\beta)$

But β is least upper bound of $\{x \in \mathbb{R} : F(x) \geq \frac{1}{2}\}$, so if $F(m) = F(\beta)$ then β is not actually an upper bound, a contradiction. So we must have $F(x) > F(\beta) \geq \frac{1}{2}$.

Assume $\lim_{y \uparrow m} F(y) \leq \frac{1}{2}$, where $m > \beta$.

Take a sequence $(y_i)_{i \in \mathbb{N}}$ such that as $i \rightarrow \infty$, $y_i \uparrow m$

In particular, we can take $\beta < y_i < m$ for all i .
 But $\beta < y_i$ means $\frac{1}{2} \leq F(\beta) < F(y_i)$, so $(y_i)_{i \in \mathbb{N}}$
 must converge to a number $> \frac{1}{2}$, so
 $\lim_{y \uparrow m} F(y) \leq \frac{1}{2}$ cannot hold. So $m > \beta$ cannot
 be a median.

(d) Show that $F(m) = \frac{1}{2}$ for any median m if F is continuous.

4.2. We have

$$\lim_{y \uparrow m} F(y) \leq \frac{1}{2} \leq F(m)$$

If F is continuous, then $\lim_{y \uparrow m} F(y) = F(m)$. So
 we have

$$F(m) \leq \frac{1}{2} \quad \text{and} \quad \frac{1}{2} \leq F(m).$$

So in fact, $F(m) = \frac{1}{2}$.